

量子與統計

2021.10.20

量子力學的矩陣形式與表象理論

Heisenberg, Born, Jordan 的矩陣力學, Schrödinger 波動力學...

厄米矩陣, $A = A^\dagger$, $A_{mn} = (A_{nm})^*$

坐標表象 $\psi(x, t)$, 動量表象 $c(p, t)$

$$\psi(x, t) = \int c(p, t) \varphi_p(x) dp, \quad \varphi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{p}{\hbar}x}$$

$$c(p, t) = \int \psi(x, t) \varphi_p^*(x) dx$$

$$\left[\int \psi(x, t) \varphi_{p'}^*(x) dx = \int \int c(p, t) \varphi_{p'}^*(x) dp \cdot \varphi_p(x) dx \right]$$

$$= \int c(p', t) \delta(p - p') dp' = c(p, t)$$

$$\int |\psi(x, t)|^2 dx = \iint \int c(p', t) \varphi_{p'}^*(x) dp' \cdot \overline{c(p, t)} \varphi_p(x) dp dx = \iint c(p', t) \overline{c(p, t)} \delta(p' - p) dp' dp$$

$$= \int c^*(p, t) c(p, t) dp$$

$$= \int |c(p, t)|^2 dp = 1$$

Parseval Equation

力學量 $\hat{Q}$, $\hat{Q}\mu_i = q_i \mu_i$

$\psi = \sum a_n(t) \mu_n(x)$, $\{a_n(t); n=1, 2, \dots\}$ 稱為 $\hat{Q}$ 表象中的波函數

$$a_n(t) = \int \psi(x, t) \mu_n^*(x) dx$$

$1 = \int \psi^* \psi = \sum |a_n|^2$, 定義 $\vec{\psi}(t) = \begin{bmatrix} a_1(t) \\ a_2(t) \\ \vdots \\ a_n(t) \end{bmatrix}$ $\hat{Q}$ 表象的態向量。

連續譜, $\psi(x, t) = \sum_n a_n(t) \psi_n(x) + \int a_g(t) \psi_g(x) dg$

$$\vec{\psi}(t) = \begin{pmatrix} a_1(t) \\ a_2(t) \\ \vdots \\ a_n(t) \\ a_g(t) \end{pmatrix}$$

坐標的本征函數 $\hat{x} \cdot \hat{x} \delta(x - x') = x' \delta(x - x') \Rightarrow x \delta(x - x') = x' \delta(x - x')$

動量的本征函數 $\varphi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{p}{\hbar}x}$

自由粒子: $\psi_p'(x, t) = \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{p}{\hbar}x - i\frac{E}{\hbar}t}$

$$c(p, t) = \int \psi_p'(x, t) \varphi_{p'}^*(x) dx = e^{-i\frac{E}{\hbar}t} \delta(p - p')$$

$$c(p) = \delta(p - p')$$

動量 $\hat{p} \delta(p - p') = p' \delta(p - p')$ v.s. $\hat{p} \psi(x, t) = p \psi(x, t)$ 坐標

$$\hat{x} \delta(x - x') = x' \delta(x - x')$$

算符的矩阵表示

坐标表象 $\hat{F}(x, -i\hbar \frac{\partial}{\partial x})$

$$\Phi(x, t) = \hat{F}(x, -i\hbar \frac{\partial}{\partial x}) \psi(x, t)$$

在表象 $\Phi(x, t) = \sum_n b_n(t) u_n(x)$

$$\psi(x, t) = \sum_n a_n(t) u_n(x)$$

$$\sum_n b_n(t) u_n(x) = \sum_n a_n(t) \hat{F}(x, -i\hbar \frac{\partial}{\partial x}) u_n(x)$$

左乘 $u_m^*(x)$ 积分

L.H.S: $b_m(t)$

$\checkmark (u_m, \hat{F} u_n)$

$$R.H.S = \sum_n \underbrace{\int u_m^*(x) \hat{F}(x, -i\hbar \frac{\partial}{\partial x}) u_n(x) dx}_{F_{mn}} \cdot a_n(t)$$

$$= \sum_n F_{mn} a_n(t)$$

$$\Rightarrow \Phi(t) = F \psi(t), \quad F = \{F_{mn}\}$$

证: $F = F^\dagger$ $\hat{F}$ 是厄米算符

$$\begin{aligned} F_{mn} &= \int u_m^*(x) \hat{F} u_n(x) dx = \int (\hat{F} u_m(x))^* u_n(x) dx \\ &= \left[\int u_n^*(x) \hat{F} u_m(x) dx \right]^* \\ &= F_{nm}^* \end{aligned}$$

证: 力学量在自身表象中为对角的厄米矩阵

$$F_{mn} = \int u_m^* \hat{F} u_n dx$$

$$Q_{mn} = \int u_m^* Q u_n dx = Q_n \delta_{mn}$$

坐标表象中 $F_{xx} = \int \delta(x'-x'') \hat{F}(x'', -i\hbar \frac{\partial}{\partial x''}) \delta(x-x') dx''$

$$= \hat{F}(x', -i\hbar \frac{\partial}{\partial x'}) \delta(x'-x')$$

$$= \hat{F}(x, -i\hbar \frac{\partial}{\partial x}) \delta(x-x')$$

表象变换

在表象 $\psi(x, t) = \sum_n a_{F,n}(t) \psi_{F,n}(x)$

在表象 $\psi(x, t) = \sum_n a_{F',n}(t) \psi_{F',n}(x) = \sum_n a_{F,n}(t) \psi_{F,n}(x)$

$$\begin{aligned} a_{F',m}(t) &= \sum_n \int \psi_{F',m}^*(x) \psi_{F,n}(x) dx \cdot a_{F,n}(t) \\ &= S_{mn} a_{F,n}(t) \end{aligned}$$

$$\vec{a}_{F'}(t) = S \vec{a}_F(t)$$

正交矩阵 $SS^\dagger = S^\dagger S = I$

力学量变换

$$\begin{cases} \vec{b}_F = F \vec{a}_F \\ \vec{b}_{F'} = F' \vec{a}_{F'} \Rightarrow \vec{b}_F = F S \vec{a}_F \Rightarrow \vec{b}_{F'} = F' S \vec{a}_F = F' S F^{-1} \vec{b}_F = F' S F^{-1} \vec{b}_F = S \vec{b}_F \\ \vec{a}_{F'} = S \vec{a}_F \end{cases}$$

~~S =~~

$$\vec{b}_{F'} = F' \vec{a}_{F'}$$

$$\vec{b}_{F'} = S \vec{b}_F = S F \vec{a}_F = S F S^{-1} \vec{b}_{F'}$$

$$\Rightarrow F' = S F S^{-1} = S F S^{\dagger}$$

證: 在 F 中.

$$L = \{L_{mn}\}, L_{mn} = (u_m(x), u_n(x))$$

在 F' 中.

$$L' = \{L'_{mn}\}, L'_{mn} = (u'_m(x), u'_n(x))$$

$$S = \{S_{mn}\}, S_{mn} = (u'_m(x), u_n(x))$$

$$\vec{S}_n^{\dagger} \vec{S}_{n'} = \sum_m S_{mn}^* S_{mn} = \sum_m (u'_m, u_n)(u'_m, u_n)$$

$$\vec{S}_n^{\dagger} \vec{S}_{n'} = \sum_m (u'_m, u_n)^* (u'_m, u_n) = \sum_m (u_n, u'_m)(u'_m, u_n)$$

$$\vec{S}_n^{\dagger} \vec{S}_{n'} = \delta_{nn}$$

$$\text{令 } \vec{S}_n^{\dagger} = (S_{n1}^*, S_{n2}^*, \dots)$$

$$\vec{Q}^{\dagger}(t) \vec{S}_{n1}^{\dagger} \vec{S}_n \vec{Q}(t) = \vec{Q}_{n1}^{\dagger}(t) \vec{Q}_n(t) = \delta_{nn}$$

量子统计

20.11.10.22

坐落表象	离散表象	力学量的平均值
态 $\psi(x, t)$	$\psi(t)$	$F = \int \psi^* F \psi dx = \psi^\dagger(t) F \psi(t)$
算符 $\hat{F}(x, -i\hbar \frac{d}{dx})$	$\{F_{mn}\}$	
算符作用到态	$\hat{F}\psi$	$F\psi(t)$
内积 $\int \Phi^*(x, t) \psi(x, t) dx$	$\Phi^\dagger(t) \psi(t)$	本征方程 $F\psi(t) = \lambda \psi(t)$

Schrödinger 方程

$$i\hbar \frac{\partial}{\partial t} \psi = \hat{H} \psi, \quad \psi = \sum_n a_n(t) u_n(x)$$

$$(u_m(x), i\hbar \frac{\partial}{\partial t} \psi) = (u_m(x), \hat{H} \psi)$$

$$\Leftrightarrow i\hbar \frac{\partial a_m(t)}{\partial t} = \sum_n H_{mn} a_n(t)$$

Dirac 符号

左矢 $\langle 1 |, \langle \psi |, \langle x |, \langle n | \dots$

右矢 $|1\rangle, |\psi\rangle, |x\rangle, |n\rangle$

$$|\psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle, \quad \langle \psi | = c_1^* \langle \psi_1 | + c_2^* \langle \psi_2 |$$

$$\langle \psi | \Phi \rangle = (\psi, \Phi)$$

$$\text{算符: } \langle \psi | \hat{A} | \Phi \rangle = \langle \psi | [\hat{A} | \Phi \rangle] = [\langle \psi | \hat{A}] | \Phi \rangle$$

平均值: $\langle \psi | \hat{A} | \psi \rangle$

$$\text{投影算符 } P_\gamma = |\gamma\rangle \langle \gamma|, \quad P_\gamma |\psi\rangle = |\gamma\rangle \langle \gamma | \psi \rangle = a_\gamma |\gamma\rangle$$

$$\sum_\gamma |\gamma\rangle \langle \gamma| = I$$

右矢 $|\psi\rangle$ 的共轭左矢是 $\langle \psi |$

Dirac 方程

$$\text{本征方程 } \hat{A} |n\rangle = a_n |n\rangle$$

$$\text{正交归一 } \langle m | n \rangle = \delta_{mn}$$

$$\text{封闭性 } \sum_n |n\rangle \langle n| = I$$

$\leftarrow u_n(x)$ 的 Dirac symbol

连续谱

$$\hat{A} |\lambda\rangle = \lambda |\lambda\rangle$$

$$\langle \lambda' | \lambda \rangle = \delta(\lambda' - \lambda)$$

$$\int_{-\infty}^{\infty} \lambda |\lambda\rangle \langle \lambda| d\lambda = I$$

连续谱本征函数 $\langle x | x_0 \rangle = \delta(x - x_0)$

$$\text{动量 } \langle x | p \rangle = \frac{1}{\sqrt{2\pi}} e^{ipx}$$

在坐标表象. 坐标 x 的延辟算符

$$\langle x' | \hat{x} | x'' \rangle$$

$$F_{mn} = (u_m, \hat{F} u_n) = \langle u_m | \hat{F} | u_n \rangle$$

Schrödinger 在不同表象下的表示

$$-i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

$$-i\hbar \frac{\partial}{\partial t} \langle x | \psi(t) \rangle = -i\hbar \frac{\partial}{\partial t} \psi(x,t)$$

$$\langle x | \hat{H} | \psi(t) \rangle$$

量子統計 [第四次作業]

2021. 10. 26

1. 2. (4.2) $\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$, $E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}$, $n=1, 2, 3, \dots$

1) 坐標, $X_{mn} = (\psi_m, \hat{x} \psi_n)$

$$\begin{aligned} &= \int_0^a x \cdot \sqrt{\frac{2}{a}} \sin\left(\frac{m\pi}{a}x\right) \cdot \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) dx \\ &= \int_0^a x \cdot \frac{1}{a} \left[\cos\left(\frac{(m-n)\pi}{a}x\right) - \cos\left(\frac{(m+n)\pi}{a}x\right) \right] dx \\ &= \int_0^a x \cdot d\left[\frac{1}{(m-n)\pi} \sin\left(\frac{(m-n)\pi}{a}x\right) - \frac{1}{(m+n)\pi} \sin\left(\frac{(m+n)\pi}{a}x\right) \right] \\ &= \frac{1}{(m+n)\pi} \int_0^a \sin\left(\frac{(m+n)\pi}{a}x\right) dx - \frac{1}{(m-n)\pi} \int_0^a \sin\left(\frac{(m-n)\pi}{a}x\right) dx \\ &= \frac{a}{(m+n)\pi^2} [1 - \cos(\pi)] - \frac{a}{(m-n)\pi^2} [1 - (-1)] \end{aligned}$$

$X_{mm} = (\psi_m, \hat{x} \psi_m)$

$$\begin{aligned} &= \int_0^a x \cdot \frac{1}{a} [1 - \cos\left(\frac{2m\pi}{a}x\right)] dx \\ &= \frac{1}{2}a - \int_0^a x \cdot \frac{1}{2m\pi} d\left(\sin\left(\frac{2m\pi}{a}x\right)\right) \\ &= \frac{1}{2}a + \frac{1}{2m\pi} \int_0^a \sin\left(\frac{2m\pi}{a}x\right) dx \\ &= \frac{1}{2}a + \frac{a}{(2m\pi)^2} (1-1) = \frac{1}{2}a \end{aligned}$$

2) 動量, $P_{mn} = (\psi_m, \hat{p} \psi_n) = -i\hbar (\psi_m, \frac{d\psi_n}{dx})$

$$\begin{aligned} &= -i\hbar \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{m\pi}{a}x\right) \cdot \sqrt{\frac{2}{a}} \cdot \frac{n\pi}{a} \cos\left(\frac{n\pi}{a}x\right) dx \\ &= -i\hbar \cdot \frac{n\pi}{a^2} \int_0^a \left[\sin\left(\frac{(m+n)\pi}{a}x\right) + \sin\left(\frac{(m-n)\pi}{a}x\right) \right] dx \\ &= -i\hbar \cdot \frac{n\pi}{a^2} \left(\frac{1}{(m+n)\pi} + \frac{1}{(m-n)\pi} \right) [1 - \cos(\pi)] \\ &= -i\hbar \cdot \frac{2mn}{a} \frac{1}{m^2 - n^2} [1 - (-1)] \end{aligned}$$

$P_{mm} = (\psi_m, \hat{p} \psi_m) = -i\hbar (\psi_m, \frac{d\psi_m}{dx})$

$$\begin{aligned} &= -i\hbar \int_0^a \psi_m d\psi_m = -\frac{i\hbar}{2} \int_0^a d(\psi_m^2) \\ &= -\frac{i\hbar}{2} [\psi_m^2(a) - \psi_m^2(0)] = 0 \end{aligned}$$

$$3. (4.5) \begin{vmatrix} 0 - \lambda_x & 1 & 0 \\ 1 & 0 - \lambda_x & 1 \\ 0 & 1 & 0 - \lambda_x \end{vmatrix} = -\lambda_x (\lambda_x^2 - 1) - (-\lambda_x) = 0$$

$$\Leftrightarrow \lambda_x (\lambda_x - \sqrt{2}) (\lambda_x + \sqrt{2}) = 0$$

$$\Leftrightarrow \lambda_x = -\sqrt{2}, 0, \sqrt{2}$$

(1) 当 $\lambda_x = 0$, 对应的特征向量是 $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$, 本征函数是 $\frac{1}{\sqrt{2}} Y_{1,-1} - \frac{1}{\sqrt{2}} Y_{1,1}$, 本征值 $\lambda_x = 0$

(2) 当 $\lambda_x = \sqrt{2}$, 对应的特征向量是 $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$, 本征函数是 $\frac{1}{2} (Y_{1,-1} + Y_{1,1}) + \frac{\sqrt{2}}{2} Y_{1,0}$, 本征值是 λ

(3) 当 $\lambda_x = -\sqrt{2}$, 对应的特征向量是 $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$, 本征函数是 $\frac{1}{2} (Y_{1,-1} + Y_{1,1}) - \frac{\sqrt{2}}{2} Y_{1,0}$, 本征值是 $-\lambda$

$$L_x = \frac{\hbar\sqrt{2}}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \frac{\hbar\sqrt{2}}{2} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \sqrt{2} & & \\ & 0 & \\ & & -\sqrt{2} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{\sqrt{2}}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{\sqrt{2}}{2} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\begin{vmatrix} 0 - \lambda_y & -i & 0 \\ i & 0 - \lambda_y & -i \\ 0 & i & 0 - \lambda_y \end{vmatrix} = -\lambda_y (\lambda_y^2 - 1) + i \cdot (-\lambda_y \cdot i) = 0$$

$$\Leftrightarrow \lambda_y (\lambda_y^2 - 1) - \lambda_y = 0$$

$$\Leftrightarrow \lambda_y = 0, -\sqrt{2}, \sqrt{2}$$

(1) 当 $\lambda_y = 0$ 时, 特征向量是 $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$, 本征函数 $\frac{1}{\sqrt{2}} (Y_{1,-1} + Y_{1,1})$, 本征值 $\lambda_y = 0$

(2) 当 $\lambda_y = \sqrt{2}$ 时, 特征向量是 $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{\sqrt{2}}{2} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$, 本征函数 $\frac{1}{\sqrt{2}} (Y_{1,-1} - Y_{1,1}) + \frac{\hbar}{\sqrt{2}} Y_{1,0}$, 本征值 $\lambda_y = \hbar$

(3) 当 $\lambda_y = -\sqrt{2}$, 特征向量是 $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{\sqrt{2}}{2} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$, 本征函数 $\frac{1}{\sqrt{2}} (Y_{1,-1} - Y_{1,1}) - \frac{\hbar}{\sqrt{2}} Y_{1,0}$, 本征值 $\lambda_y = -\hbar$

$$L_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \hbar & & \\ & 0 & \\ & & -\hbar \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{\hbar}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{\hbar}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

4. (补充题 1) Hermitian 算符 $\hat{A}, \hat{B}$. $\hat{A}^2 = \hat{B}^2 = \hat{I}$. $\hat{A}\hat{B} + \hat{B}\hat{A} = 0$

(1) 设 $\hat{A}$ 的本征函数是 u_n

$$\hat{A}u_n = \lambda u_n$$

$$\Rightarrow \hat{A}^2 u_n = \lambda^2 u_n = \hat{I} u_n = u_n$$

$$\Rightarrow \lambda^2 = 1 \Leftrightarrow \lambda = \pm 1$$

$$\text{且 } \hat{A}(\hat{B}u_n) = \hat{B}(-\hat{A}u_n) = -\lambda(\hat{B}u_n)$$

因此 $\hat{A}$ 的本征值是 ± 1 . 对应的本征函数是 u_n 和 $\hat{B}u_n$

若本征值为 1 的本征函数是 $u_{A,1}$, 则本征值为 -1 的本征函数是 $\hat{B}u_{A,1} = u_{A,2}$

因此 $\hat{B}$ 的本征值为 1 的本征函数是 $\hat{B}u_{A,1}$

~~解法为 $\hat{B}u_{A,1} = u_{A,2}$~~

$$A = \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$(2) \begin{vmatrix} 0-\lambda & 1 \\ 1 & 0-\lambda \end{vmatrix} = \lambda^2 - 1 = 0, B \text{ 的本征值是 } \pm 1$$

$$\lambda_B = 1 \text{ 时, 本征函数是 } \psi_{B,1} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\lambda_B = -1 \text{ 时, 本征函数是 } \psi_{B,2} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

5. (补充题 2) (1) 在坐标 x 表象中.

$$F_{xx} = \int \delta(x''-x') x'' \delta(x''-x) dx''$$

$$= x' \delta(x'-x)$$

$$\langle x' | \hat{p}_x^2 | x'' \rangle = \int \langle x' | p' \rangle \langle p' | \hat{p} | p'' \rangle \langle p'' | x'' \rangle dp' dp''$$

$$= \int \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{p'}{\hbar}x'} p' \delta(p'-p'') \frac{1}{\sqrt{2\pi\hbar}} e^{-i\frac{p''}{\hbar}x''} dp' dp''$$

$$= \frac{1}{2\pi\hbar} \int p' e^{i\frac{p'}{\hbar}(x'-x'')} dp'$$

$$= \frac{1}{2\pi\hbar} (-i\hbar \frac{\partial}{\partial x'}) \int e^{i\frac{p'}{\hbar}(x'-x'')} dp'$$

$$= -i\hbar \frac{\partial}{\partial x'} \delta(x'-x'')$$

$$\langle x' | \hat{H} | x'' \rangle = \int \delta(x'-x) \cdot \left(\frac{p^2}{2m} + V(x) \right) \delta(x-x'') dx$$

$$= \int \delta(x-x') \cdot \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \delta(x-x'') dx$$

$$= -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x'^2} \delta(x'-x'') + V(x') \delta(x'-x'')$$

$$= \iint \frac{1}{\sqrt{2\pi\hbar}} e^{-i\frac{p_x'}{\hbar}x'} \cdot x' \delta(x'-x'') \cdot \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{p_x''}{\hbar}x'} dx' dx''$$

$$(b) \langle p_x' | \hat{x} | p_x'' \rangle = \iint \langle p_x' | x' \rangle \langle x' | \hat{x} | x'' \rangle \langle x'' | p_x'' \rangle dx' dx''$$

$$= \iint \frac{1}{\sqrt{2\pi\hbar}} e^{-i\frac{p_x'}{\hbar}x'} \cdot x' \delta(x'-x'') \cdot \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{p_x''}{\hbar}x''} dx' dx''$$

$$= \frac{1}{2\pi\hbar} \int e^{i\frac{(p_x''-p_x')x'}{\hbar}} \cdot x' dx' = \frac{1}{2\pi\hbar} (i\hbar \frac{\partial}{\partial p_x'}) \cdot \int e^{i\frac{(p_x''-p_x')x'}{\hbar}} dx'$$

$$= \frac{1}{2\pi\hbar} \frac{\partial}{\partial p_x'} \int e^{i\frac{(p_x''-p_x')x'}{\hbar}} dx' = i\hbar \frac{\partial}{\partial p_x'} \delta(p_x''-p_x')$$

$$\langle p_x' | \hat{p}_x | p_x'' \rangle = p_x' \delta(p_x'-p_x'')$$

$$\langle p_x' | \hat{H} | p_x'' \rangle = \int \frac{1}{\sqrt{2\pi\hbar}} e^{-i\frac{p_x'}{\hbar}x} \cdot \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \cdot \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{p_x''}{\hbar}x} dx$$

$$= \frac{1}{2\pi\hbar} \int V(x) e^{i\frac{(p_x''-p_x')x}{\hbar}} dx$$

$$\langle p_x' | \hat{H} | p_x'' \rangle = \int \frac{1}{\sqrt{2\pi\hbar}} e^{-i\frac{p_x'}{\hbar}x} \cdot \left(\frac{\hat{p}_x^2}{2m} + V(x) \right) \cdot \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{p_x''}{\hbar}x} dx$$

$$= \int \frac{1}{\sqrt{2\pi\hbar}} e^{-i\frac{p_x'}{\hbar}x} \cdot \left[\frac{p_x''^2}{2m} + V(x) \right] \cdot \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{p_x''}{\hbar}x} dx$$

$$= \frac{p_x''^2}{2m} \delta(p_x'-p_x'') + \frac{1}{2\pi\hbar} \int V(x) \cdot e^{i\frac{(p_x''-p_x')x}{\hbar}} dx$$

6. (例 2. 题 第一致)